# Microscopic quantum mechanisms of stabilization of relativistic electroweak objects with critical mass 2.8 M☉

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## Abstract

This article presents a new theoretical model describing final stages of gravitational collapse of stellar cores with mass 2.8 M☉. Proposed mechanism, in which extreme energy density induces reverse phase transition of Higgs field (restored symmetry, v = 0), leads to loss of mass acquired by elementary particles. It is shown that dimensionless structural parameter η becomes independent of macroscopic mass of object and gravitational constant. Additionally, we analyze dual pressure: of dimensionless electroweak radiation and electrostatic generated by tunneling of neutrons to protons and beta decay activation. Including relativistic spin of matter reaching speed of light leads to modification of local spacetime geometry, eliminating formation of singularities and event horizon. The final radius of stable electroweak core is estimated at approx. 15.8 meters.

## 1. Introduction

Classical general theory of relativity (GTR) predicts that collapse of matter with mass exceeding Tolman-Oppenheimer-Volkoff (TOV) limit inevitably leads to formation of event horizon and point singularity of infinite density of matter and spacetime curvature. This result is commonly regarded as manifestation of limitation of classical description and signature of necessity of introducing quantum effects.

In present work we abandon approach based on searching for complete quantum theory of gravity for thing of analysis of behavior of matter of Standard Model in extreme energy densities, such as accompanying collapse of core with mass 2.8 M☉. We show that before matter reaches Planck scale, physics of phase transitions of electroweak interactions triggers universal microscopic defense mechanism, capable of completely stopping gravitational collapse.

## 2. Package Formula and independence from macroscopic scale

Consider collapsing core with mass M = 2.8 M☉. Define radius of electroweak core R_core as scale at which local energy density corresponds to energy of Higgs scale E_H:

$$R_{core} = G M m_0 / E_H \quad (1)$$

where G is gravitational constant, $m_0$ is rest mass of reduced nuclear particle (nucleon), and E_H = 246 GeV represents vacuum expectation value of Higgs field. Mass density of core ρ_core and associated relativistic radiation pressure p_core are defined by classical relations:

$$\rho_{core} = M / R^3_{core} \quad (2)$$
$$p_{core} = 1/3\ \rho_{core} c^2 = 1/3\ M/R^3_{core} c^2 \quad (3)$$

In order to study preservation of structure we introduce dimensionless parameter η, describing ratio of internal pressure energy to macroscopic gravitational energy in volume of core:

$$\eta = p_{core} R^4_{core} / G M^2 \quad (4)$$

Substituting explicit expression for pressure (Equation 3) to definition of parameter η, we obtain:

$$\eta = [1/3\ M/R^3_{core} c^2 \times R^4_{core}] / G M^2 = 1/3\ M c^2 R_{core} / G M^2 = 1/3\ c^2 R_{core} / G M \quad (5)$$

In crucial step we derive output expression for radius of core R_core (Equation 1) to above result:

$$\eta = 1/3\ c^2/GM \times G M m_0 / E_H = 1/3\ c^2 m_0 / E_H \quad (6)$$

Obtained result has fundamental significance: structural parameter η is completely independent of mass M and gravitational constant G. This means that resistance of matter against gravity in this regime does not depend on macroscopic scale of phenomenon, but is dictated by universal microscopic constants of Standard Model.

## 3. Reverse Higgs phase transition and dualism of pressures

When collapsing core reaches energy density of order of electroweak scale, two fundamental quantum processes occur in parallel:

1. Restoration of Higgs symmetry (v = 0): Extreme temperature and energy density cause melting of W± and Z bosons (~160-246 GeV). Higgs field transitions to state of zero vacuum expectation value. Consequently mechanism generating rest masses of elementary particles ceases to operate. Particles become massless and move with speed of light (c), generating powerful, isotropic relativistic radiation pressure captured by coefficient 1/3 in equation of state.

2. Activation of electrostatic forces: Extreme packing of matter induces phenomenon of quantum tunneling of neutrons to protons. Triggered is violent process of beta decay:

$$n \leftrightarrow p + e^- + \bar{\nu}_e \quad (7)$$

Appearance of macroscopic density of positive charge (protons) generates gigantic electrostatic repulsion (Coulomb). This force, being many orders of magnitude stronger than gravitational interaction at small distances, creates second, complementary vector of pressure expanding core.

## 4. Relativistic spin and spacetime geometry

Introduction to model of momentum (spin) of particles moving with speed of light radially changes structure of geometry of object. According to Kerr solution for rotating masses, centrifugal force counteracts spherical collapse.

When rotation speed of collapsing matter particles reaches relativistic limit (c), classical singularity point undergoes stretching in two-dimensional ring, and effect of dragging of inertial frames (frame-dragging) reaches critical value. Extreme spin modifies Einstein curvature tensor in such a way that it prevents formation of standard event horizon. Quantum implementation of relativistic spin causes that instead of black hole, stable relativistic vortex of spacetime is created, in which gravity is permanently balanced by microscopic dynamic effects.

## 5. Numerical analysis of object scale

Using derived Equation 1, we can accurately determine physical size of created electroweak object for mass M = 2.8 M☉. Assume standard values of constants in SI system:

- $G = 6.674 \times 10^{-11}\ m^3\ kg^{-1}\ s^{-2}$
- $M = 2.8 \times 1.989 \times 10^{30}\ kg \approx 5.569 \times 10^{30}\ kg$
- $m_0 = 1.675 \times 10^{-27}\ kg$
- $E_H = 246\ GeV = 246 \times 10^9 \times 1.602 \times 10^{-19}\ J \approx 3.941 \times 10^{-8}\ J$

Substituting above data into formula:

$$R_{core} = (6.674 \times 10^{-11}) \times (5.569 \times 10^{30}) \times (1.675 \times 10^{-27}) / 3.941 \times 10^{-8} \approx 15.8\ m \quad (8)$$

Object with mass almost three Solar masses reaches equilibrium at radius amounting to only 15.8 meters (diameter approx. 31.6 m). For comparison, classical Schwarzschild radius for same mass is $R_S \approx 8.3$ km. Since $R_{core} \ll R_S$, universal quantum-spin mechanism redefines interior metric before singularity formation occurs.

## 6. Conclusions

Presented model proves that inclusion of reverse Higgs phase transition, beta decay and relativistic spin allows elimination of gravitational singularities within known mechanisms of Standard Model and GTR. Key feature of model - complete independence of parameter of preservation η from gravity and mass - indicates that at threshold of black hole formation, quantum physics takes over full control over spacetime geometry. Created electroweak object constitutes new, stable form of macroscopic quantum matter.

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